

		time
16	C	External force required to push mass up the slope at constant speed, $F = mg \sin \theta + f_r$ Mean power $\langle P \rangle = Fv = (mg \sin \theta + f_r) \left(\frac{d}{t} \right)$ $= \left[(4.00)(9.81) \sin 30^\circ + 8.00 \right] \left(\frac{6.00}{3.00} \right)$ $= 55.24 \approx 55.2 \text{ W}$
17	C	By conservation of energy